C PROGRAMMING LAB RECORD

INDEX

Expt. No.	Name of the Experiment	Page No.
1	C program to print the size of different data types	2
2	Perimeter and area of a triangle	3
3	Area and circumference of a circle	4
4	Convert Fahrenheit to Celsius	5
5	Simple interest	6
6	Swap two numbers without using a temporary variable	7
7	Swap two numbers using a temporary variable	8
8	Check a > b using the conditional operator	9
9	Arithmetic and relational operations using short-name method	10
10	Evaluate the expression (a+b)*c	11
11	Equivalence of two numbers	12
12	Largest of two numbers	13
13	Maximum of three numbers	14
14	Smallest of two numbers	15
15	Energy bill using else-if ladder	16
16	Leap year or not	17
17	Odd or even	18
18	Gross salary of an employee	19
19	Average of six subjects with division classification	20
20	Day of a week using switch case	22
21	Menu driven arithmetic calculations using switch case	23

Expt. No. : 1	Date : .....

Aim :

To write a C program to print the size of different data types.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    clrscr();
    printf(" SIZE OF DATA TYPES : \n\n");
    printf(" Size of int = %d bytes\n", sizeof(int));
    printf(" Size of char = %d bytes\n", sizeof(char));
    printf(" Size of float = %d bytes\n", sizeof(float));
    printf(" Size of double = %d bytes\n", sizeof(double));
    getch();
}
```

Teacher's Signature

Expt. No. : 2	Date : .....

Aim :

To find the perimeter and area of a triangle.

Program :

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
    float a, b, c, s, area, perimeter;
    clrscr();

    printf(" Enter three sides of triangle : ");
    scanf("%f %f %f", &a, &b, &c);

    perimeter = a + b + c;
    s = perimeter / 2;
    area = sqrt(s * (s - a) * (s - b) * (s - c));

    printf(" Perimeter = %f", perimeter);
    printf("\n Area = %f", area);

    getch();
}
```

Teacher's Signature _____

Expt. No. : 3	Date : .....

Aim :

To find the area and circumference of a circle.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    float r, area, circumference;
    clrscr();

    printf(" Enter radius : ");
    scanf("%f", &r);

    area = 3.14 * r * r;
    circumference = 2 * 3.14 * r;

    printf(" Area = %f", area);
    printf("\n Circumference = %f", circumference);

    getch();
}
```

Teacher's Signature _____

Expt. No. : 4	Date : .....

Aim :

To convert temperature in Fahrenheit to Celsius.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    float f, c;
    clrscr();

    printf(" Enter temperature in Fahrenheit : ");
    scanf("%f", &f);

    c = (f - 32) * 5 / 9;

    printf(" Temperature in Celsius = %f", c);

    getch();
}
```

Teacher's Signature

Expt. No. : 5	Date : .....

Aim :

To find the Simple Interest.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    float p, r, t, si;
    clrscr();

    printf(" Enter principal, rate and time : ");
    scanf("%f %f %f", &p, &r, &t);

    si = (p * r * t) / 100;

    printf(" Simple Interest = %f", si);

    getch();
}
```

Teacher's Signature

Expt. No. : 6	Date : .....

Aim :

To swap two numbers without using a temporary variable.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    a = a + b;
    b = a - b;
    a = a - b;

    printf(" After swapping : a = %d, b = %d", a, b);

    getch();
}
```

Teacher's Signature

Expt. No. : 7	Date : .....

Aim :

To swap two numbers using a temporary variable.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b, temp;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    temp = a;
    a = b;
    b = temp;

    printf(" After swapping : a = %d, b = %d", a, b);

    getch();
}
```

Teacher's Signature

Expt. No. : 8	Date : .....

Aim :

To input two numbers a and b and check a > b using the conditional operator.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    (a > b) ? printf("a is greater") : printf("b is greater or equal");

    getch();
}
```

Teacher's Signature

Expt. No. : 9	Date : .....

Aim :

To perform arithmetic and relational operations using the short-hand method.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b, x;
    clrscr();
    printf(" Enter two values : ");
    scanf("%d %d", &a, &b);

    printf("\n Arithmetic operations : ");

    x = a;
    x += b;
    printf("\n Addition = %d", x);

    x = a;
    x -= b;
    printf("\n Subtraction = %d", x);

    x = a;
    x *= b;
    printf("\n Multiplication = %d", x);

    x = a;
    x /= b;
    printf("\n Division = %d", x);

    printf("\n\n Relational operators : ");
    printf("\n a>b = %d", a > b);
    printf("\n a<b = %d", a < b);
    printf("\n a==b = %d", a == b);
    printf("\n a!=b = %d", a != b);
    printf("\n a>=b = %d", a >= b);
    printf("\n a<=b = %d", a <= b);

    getch();
}
```

Teacher's Signature

Expt. No. : 10	Date : .....

Aim :

To evaluate the expression (a + b) * c.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b, c, result;
    clrscr();

    printf(" Enter a, b and c : ");
    scanf("%d %d %d", &a, &b, &c);

    result = (a + b) * c;

    printf(" Result = %d", result);

    getch();
}
```

Teacher's Signature _____

Expt. No. : 11	Date : .....

Aim :

To check the equivalence of two numbers.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two integers a and b : ");
    scanf("%d %d", &a, &b);

    if (a == b)
        printf("\n 1. Condition (a==b) is TRUE");
    else
        printf("\n 1. Condition (a==b) is FALSE");

    if (a - b == 0)
        printf("\n 2. Condition (a-b==0) is TRUE");
    else
        printf("\n 2. Condition (a-b==0) is FALSE");

    if (b != 0)
    {
        if (a / b == 1)
            printf("\n 3. Condition (a/b==1) is TRUE");
        else
            printf("\n 3. Condition (a/b==1) is FALSE");
    }
    else
        printf("\n 3. Division by zero is not possible");

    getch();
}
```

Teacher's Signature

Expt. No. : 12	Date : .....

Aim :

To find the largest of two numbers.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    if (a > b)
        printf(" a is larger");
    else
        printf(" b is larger");

    getch();
}
```

Teacher's Signature

Expt. No. : 13	Date : .....

Aim :

To find the maximum of three numbers.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b, c;
    clrscr();

    printf(" Enter three numbers : ");
    scanf("%d %d %d", &a, &b, &c);

    if (a > b && a > c)
        printf(" A is largest");
    else if (b > a && b > c)
        printf(" B is largest");
    else
        printf(" C is largest");

    getch();
}
```

Teacher's Signature

Expt. No. : 14	Date : .....

Aim :

To find the smallest of two numbers.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    if (a < b)
    {
        printf(" a is smaller");
    }
    else
    {
        printf(" b is smaller");
    }

    getch();
}
```

Teacher's Signature

Expt. No. : 15	Date : .....

Aim :

To calculate the energy bill by reading the starting and ending meter readings using an else-if ladder.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    float start, end, units, bill;
    clrscr();

    printf(" Enter starting meter reading : ");
    scanf("%f", &start);
    printf(" Enter ending meter reading : ");
    scanf("%f", &end);

    units = end - start;

    if (units > 200 && units <= 500)
    {
        bill = units * 2.50;
    }
    else if (units >= 100 && units < 200)
    {
        bill = units * 2.50;
    }
    else
    {
        bill = units * 1.50;
    }

    printf("\n Units consumed = %f", units);
    printf("\n Energy bill = Rs %f", bill);

    getch();
}
```

Teacher's Signature _____

Expt. No. : 16	Date : .....

Aim :

To check whether a year is a leap year or not.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int year;
    clrscr();

    printf(" Enter a year : ");
    scanf("%d", &year);

    if (year % 4 == 0)
    {
        printf(" Leap year");
    }
    else
    {
        printf(" Not a leap year");
    }

    getch();
}
```

Teacher's Signature

Expt. No. : 17	Date : .....

Aim :

To check whether a number is odd or even.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a;
    clrscr();

    printf(" Enter the number : ");
    scanf("%d", &a);

    if (a % 2 == 0)
    {
        printf(" a is even");
    }
    else
    {
        printf(" a is odd");
    }

    getch();
}
```

Teacher's Signature _____

Expt. No. : 18	Date : .....

Aim :

To calculate the gross salary of an employee.

Basic Salary	DA	HRA	Conveyance
>= 5000	110% of basic	20% of basic	500
> 3000 & < 5000	100% of basic	15% of basic	400
<= 3000	90% of basic	10% of basic	300

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    float basic_salary, DA, HRA, conveyance, gross_salary;
    clrscr();

    printf(" Enter basic salary : ");
    scanf("%f", &basic_salary);

    if (basic_salary >= 5000)
    {
        DA = 1.10 * basic_salary;
        HRA = 0.20 * basic_salary;
        conveyance = 500.00;
    }
    else if (basic_salary >= 3000 && basic_salary < 5000)
    {
        DA = 1.00 * basic_salary;
        HRA = 0.15 * basic_salary;
        conveyance = 400.00;
    }
    else if (basic_salary >= 3000)
    {
        DA = 0.90 * basic_salary;
        HRA = 0.10 * basic_salary;
        conveyance = 300.00;
    }
    else
    {
        printf(" Invalid value");
    }

    gross_salary = basic_salary + DA + HRA + conveyance;
    printf(" Gross salary = %.2f", gross_salary);

    getch();
}
```

Teacher's Signature

Expt. No. : 19	Date : .....

Aim :

To find the average of marks in six subjects and the result.

```
Average              Result
34 & < 50            Third Division
50 & < 60            Second Division
60 & < 75            First Division
75 & 100             Distinction
Mark in any subject < 35   Fail
```

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    float marks[6], sum = 0, avg;
    int i = 0, has_failed = 0;
    clrscr();

    printf(" Enter the marks for 6 subjects : \n");
    do
    {
        printf(" Subject %d : ", i + 1);
        scanf("%f", &marks[i]);
        if (marks[i] < 35.0)
        {
            has_failed = 1;
        }
        sum += marks[i];
        i++;
    } while (i < 6);

    avg = sum / 6.0;

    printf("\n ... AVERAGE RESULT ... \n");
    printf(" Average mark : %.2f\n", avg);

    printf(" Result : ");
    if (has_failed)
    {
        printf(" Fail (marks in one or more subjects are less than 35)\n");
    }
    else if (avg >= 75.0 && avg <= 100.0)
    {
        printf(" Distinction\n");
    }
    else if (avg >= 60.0 && avg < 75.0)
    {
        printf(" First Division\n");
    }
    else if (avg >= 50.0 && avg < 60.0)
    {
        printf(" Second Division\n");
```

Teacher's Signature

```
    }
    else if (avg >= 35.0 && avg < 50.0)
    {
        printf(" Third Division\n");
    }
    else
    {
        printf(" Fail\n");
    }

    getch();
}
```

Teacher's Signature _____

Expt. No. : 20	Date : .....

Aim :

To print the day of a week using switch case.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int choice;
    clrscr();

    printf(" Menu\n");
    printf(" * * * * ..... \n");
    printf(" 1. Sunday\n");
    printf(" 2. Monday\n");
    printf(" 3. Tuesday\n");
    printf(" 4. Wednesday\n");
    printf(" 5. Thursday\n");
    printf(" 6. Friday\n");
    printf(" 7. Saturday\n");
    printf(" Enter your choice : ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf(" Sunday");
            break;
        case 2:
            printf(" Monday");
            break;
        case 3:
            printf(" Tuesday");
            break;
        case 4:
            printf(" Wednesday");
            break;
        case 5:
            printf(" Thursday");
            break;
        case 6:
            printf(" Friday");
            break;
        case 7:
            printf(" Saturday");
            break;
        default:
            printf(" Invalid choice");
    }

    getch();
}
```

Teacher's Signature _____

Expt. No. : 21	Date : .....

Aim :

To write a menu driven program to perform arithmetic calculations using switch case.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int choice;
    int x, y;
    float a, b, sum, sub, mul, div;
    clrscr();

    printf(" Menu\n");
    printf(" * * * ..... * \n");
    printf(" 1. Addition\n");
    printf(" 2. Subtraction\n");
    printf(" 3. Multiplication\n");
    printf(" 4. Division\n");
    printf(" 5. Remainder\n");
    printf(" 6. Largest of two numbers\n");
    printf(" 7. Exit\n");
    printf(" Enter your choice : ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf(" Enter two numbers : ");
            scanf("%f %f", &a, &b);
            sum = a + b;
            printf(" Addition = %.2f", sum);
            break;
        case 2:
            printf(" Enter two numbers : ");
            scanf("%f %f", &a, &b);
            sub = a - b;
            printf(" Subtraction = %.2f", sub);
            break;
        case 3:
            printf(" Enter two numbers : ");
            scanf("%f %f", &a, &b);
            mul = a * b;
            printf(" Multiplication = %.2f", mul);
            break;
        case 4:
            printf(" Enter two numbers : ");
            scanf("%f %f", &a, &b);
            div = a / b;
            printf(" Division = %.2f", div);
            break;
        case 5:
```

Teacher's Signature _____

```
            printf(" Enter two numbers : ");
            scanf("%d %d", &x, &y);
            printf(" Remainder = %d", x % y);
            break;
        case 6:
            printf(" Enter two numbers : ");
            scanf("%d %d", &x, &y);
            if (x > y)
                printf(" Largest = %d", x);
            else
                printf(" Largest = %d", y);
            break;
        case 7:
            printf(" Exit");
            break;
        default:
            printf(" Invalid choice");
    }

    getch();
}
```

Teacher's Signature _____